Coursework Introduction

Yulan He

Coursework Assessment

- Task: Open Question Answering
- Dataset: HotPotQA
- Potential methodologies
- Evaluation
- Marking Criteria
- Submission

Task: Open Question-Answering

- **Open Question-Answering:**
  - "Question Answering (QA)" → The task is to find or generate an accurate answer to a natural-language question.
  - "Open-domain" → The QA system isn't limited to one particular domain (such as health or finance); it can answer any given questions.

Paragraph A: **Peter Marc Jacobson**
Peter Marc Jacobson (born October 27, 1957) is an American television writer, director and producer, and actor. He is best known as the co-creator of the popular sitcom "The Nanny", which he created and wrote with his then wife actress Fran Drescher, who was the star of the series. He was often credited as Peter Marc in his early acting roles.

Paragraph B: **Fran Drescher**
Francine Joy "Fran" Drescher (born September 30, 1957) is an American actress and activist. She is best known for her role as Fran Fine in the hit TV series "The Nanny" (1993–99), and for her nasal voice and thick New York accent.

Q: Peter Marc Jacobson is best known as the co-creator of the popular sitcom "The Nanny", which he created and wrote with his then wife an actress born in which year ?
A: 1957

Dataset

- HotPot QA dataset: https://hotpotqa.github.io/
  - Training set – use it to fine-tune your LLM
  - Testing set – not publicly available
  - Development set – use it for testing/evaluating the performance of your QA system
    - DO NOT use the development set for training or hyperparameter tuning.
    - Using development set for training will be penalised.

- How do you know if I used the development set for training?
  - Since you need to submit your code, we can test your model with some special cases.

Potential Methodologies

- Prompting and in-context learning (Week 7)
- Supervised Fine-Tuning (SFT) (Week 9)
- Retrieval-Augmented Generation (RAG) (Week 10)
- LLM reasoning-based methods (Week 11)

Evaluation

- Evaluation is done on the HotpotQA Development Set
- You need to submit your QA results following the format below:

```
{"answer":
  {
    "5ae66d515542992ae0d16346":"1927",
    "5a83aaeb5542996488c2e483":"Graduation",
    "5a8c66db554299585d9e3683":"2007",
    "5ae16ea85542990adbacf790":"31 July 1975",
    "5ac10832554299012d1db687":"several",
    "5ae2c4b8554299495565db31":"New York City",
    "5a78d38455429970f5fffd94":"Julian Assange",
    "5adce4f65542992c1e3a2474":"United States Department of Health and Human Services",
    "5ab6654455429954757d3281":"genus of flowering",
    "5adf39ce5542995534e8c76d":"Shakespeare performances throughout India",
    "5a7619765542992db9473728":"Randers, Denmark",
    "5a77a3905542992a6e59df40":"Duke Blue Devils",
    "5ac38ce255429939154137c2":"water poured over ice",
    "5a8a38135542996c9b8d5e59":"Chancellor of Germany",
    "5ac3bd0c554299204fd21e21":"filmmaker",
    "5adcaa7f5542990d50227cdb":"graduate students",
    "5a85b0c85542994c784ddb40":"Texas Tech University",
    "5add6b595542992200553b0c":"Atlanta, Georgia",
   }
}
```

Marking Criteria

- The coursework will not be graded based on model performance, but rather on the overall research quality, originality, and clarity of presentation.
- Assessment will be conducted based on three aspects:
  - Novelty
  - Organisation and Experimental Support
  - Report Writing
    - Reference: https://arxiv.org/abs/2503.01606
    - It is our research paper on this dataset, you can follow the format to organise your report.

Report Format

- The report should follow the ACL 2026 LaTeX template
- The total length of the report (excluding references) should be between 4 and 6 pages.
  - Submissions exceeding or falling short of this range will receive a penalty.
  - References section should contain at least 10 references.
- Using of generative AI:
  - **Policy:** The module follows Model 2 of the Informatics Handbook AI policy, which authorises generative AI for literature search.
  - **Writing:** AI may be used for spelling and grammar correction.
  - **Research:** AI may help gather relevant information, such as material for the literature review.
  - **Coding:** AI use is limited to refactoring routines and fixing small bugs.
  - **Acknowledgement:** Students must include a brief statement of how far the LLM tool was used, together with the prompts they gave it.

Submission

- Submission contains
  - A report
  - A zip file containing the following:
    - A readme file on Python environment setup and code execution instruction.
    - Python code.
    - Predictions on the development (Dev) set, formatted according to the Dev example provided on the HotPotQA webpage.
- Name the zip file using your team / group name (e.g., groupABC).